

| Question |     |      |  | Marking details                                                                                                                               | Marks available |          |          |          |          |          |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                               | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 6        | (a) |      |  | [Very] high temperature (1)<br>[Very] high pressure (1)<br>Don't accept heat                                                                  | 2               |          |          | 2        |          |          |
|          | (b) | (i)  |  | Line drawn at 590 nm                                                                                                                          | 1               |          |          | 1        |          |          |
|          |     | (ii) |  | Longer wavelength / red shifted / moved to the right side (1)<br>More distant star moving away from Earth (1)<br>so its wave is stretched (1) | 3               |          |          | 3        |          |          |
|          |     |      |  | <b>Question 6 total</b>                                                                                                                       | <b>6</b>        | <b>0</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>0</b> |